

PMDM–MOC Formalization: Prime-Graded Density Operators and QFT Constructions (A/B/C)

This document formalizes a **prime-graded density-operator framework** (PMDM) compatible with the **prime-indexed move** of Multiplicity Operator Calculus (MOC): each prime generates its own (generally) **noncommuting operator family** acting on both **state** and **relation** degrees of freedom of a hypergraph. Validation is handled by a **resonance functional** $R \in [0, 1]$.

The goal is to produce a formulation that is simultaneously:

- **Mathematically admissible** (density operators remain PSD and trace 1; dynamics are CPTP);
- **Noncommutation-friendly** (order-sensitive prime-indexed updates are meaningful);
- **QFT-grade** (admits path-integral and Keldysh boundary-condition realizations);
- **Neuromorphic-implementable** (naturally supports delayed updates, gating, and robust mixtures).

1. Core state space and PMDM definition

1.1 Prime register and hypergraph Hilbert space

Let $\mathcal{P}_N = \{p_1, \dots, p_N\}$ be a truncated prime set.

- **Prime register:** $\mathcal{H}_{\mathbb{P}} = \text{span}\{|p\rangle : p \in \mathcal{P}_N\} \cong \mathbb{C}^N$.
- **Hypergraph state+relation space:** $\mathcal{H}_{\text{HG}} = \mathcal{H}_V \otimes \mathcal{H}_E$, where
- $\mathcal{H}_V$ encodes node (vertex) states,
- $\mathcal{H}_E$ encodes hyperedge / relation states (e.g., adjacency tensors, relation modes, clause-like constraints).
- **Total space:** $\mathcal{H}_{\text{tot}} = \mathcal{H}_{\mathbb{P}} \otimes \mathcal{H}_{\text{HG}}$.

1.2 Prime-graded density operator (PMDM)

A **prime-graded density operator** is

$$\widetilde{\rho}(t) \in \mathcal{D}(\mathcal{H}_{\text{tot}}), \quad \widetilde{\rho} \succeq 0, \quad \text{Tr}(\widetilde{\rho}) = 1.$$

Block decomposition in the prime basis:

$$\widetilde{\rho}(t) = \sum_{p,q \in \mathcal{P}_N} |p\rangle\langle q| \otimes \rho_{pq}(t),$$

with operators $\rho_{pq}(t) \in \mathcal{B}(\mathcal{H}_{\text{HG}})$.

Two regimes:

- **Mixture / neuromorphic (diagonal):**

$$\widetilde{\rho}(t) = \sum_{p \in \mathcal{P}_N} w_p(t) |p\rangle\langle p| \otimes \rho^{(p)}(t), \quad w_p \geq 0, \sum_p w_p = 1.$$

- **Coherent / QFT-grade (off-diagonal allowed):** $\rho_{pq}(t) \neq 0$ for $p \neq q$, enabling cross-prime interference and genuinely order-sensitive operator composition.

1.3 Prime-weight maps (admissible)

Parameterize weights via logits $\eta_p \in \mathbb{R}$:

$$w_p(\eta) = \frac{e^{\eta_p}}{\sum_q e^{\eta_q}}.$$

Optional prime-shaped initialization:

- Zeta-like prior (truncated): $\eta_p(0) = -s \log p$ so $w_p \propto p^{-s}$.
- Learned prime embeddings: $\eta_p = a_p^\top \theta$ with trainable θ .

2. Prime-indexed operator families (MOC-compatible)

2.1 Controlled prime operators

For each prime p , let $\{U_{p,\alpha}\}_\alpha \subset \mathcal{B}(\mathcal{H}_{\text{HG}})$ be a (generally noncommuting) family acting on nodes+relations.

Lift them to $\mathcal{H}_{\text{tot}}$ as controlled operators:

$$\widetilde{U}_{\{p,\alpha\}} = |p\rangle\langle p| \otimes U_{\{p,\alpha\}}.$$

These are block-diagonal in the prime register, enabling clean per-prime action.

2.2 Cross-prime couplers (optional but crucial)

To avoid trivial superselection and enable cross-prime noncommutation, introduce couplers:

$$\widetilde{V} = \sum_{\{p \neq q\}} |p\rangle\langle q| \otimes V_{\{pq\}},$$

with $V_{pq} \in \mathcal{B}(\mathcal{H}_{\text{HG}})$. These generate prime-coherence and allow genuinely prime-braided dynamics.

3. Admissible dynamics (CPTP by construction)

Let $\mathcal{L}$ be a Lindbladian:

$$\frac{d\tilde{\rho}}{dt} = -i[\tilde{H}, \tilde{\rho}] + \sum_k \text{Big}(\tilde{L}_k \tilde{\rho}, \tilde{L}_k^\dagger - \frac{1}{2}(\tilde{L}_k^\dagger \tilde{L}_k + \tilde{L}_k \tilde{L}_k^\dagger))$$

Choose

$$\tilde{H} = \sum_p |p\rangle\langle p| \otimes H_p + H_{\text{couple}},$$

and similarly for $\tilde{L}_k$, possibly with controlled and cross-prime components. This guarantees PSD and trace preservation.

3.2 Split-step implementation and commutator budget (engineering constraint)

If $\mathcal{L} \approx \sum_p \mathcal{L}_p$ and one simulates by sequentially applying channels $\exp(\Delta t \mathcal{L}_p)$, the leading error is controlled by commutators $[\mathcal{L}_p, \mathcal{L}_q]$. A practical bound requirement is:

$$\Delta t \sum_{p < q} \|[\mathcal{L}_p, \mathcal{L}_q]\| \ll 1.$$

Schedulers can reduce error by ordering or grouping primes with small commutators.

Define a feature map Φ extracting prime-aware observables from $\tilde{\rho}$:

$$\Phi(\tilde{\rho}) = \text{Big}(\{\text{Tr}\{\text{HG}\}[(|p\rangle\langle p| \otimes I)\tilde{\rho}]_{-p}, \{S(\rho^{(p)})\}_{-p}, \{[A_p, A_q]\}, \dots\})$$

Let $\mathcal{T}$ denote a target or constraint specification in the same feature space.

A simple bounded resonance is:

$$R(\tilde{\rho}; \mathcal{T}) = \exp\big(-\lambda D(\Phi(\tilde{\rho}), \mathcal{T})\big) \in (0, 1],$$

where D is a metric/divergence (e.g., squared error, KL on normalized feature vectors, Wasserstein on distributions).

5. Delay-friendly learning of prime weights (neuromorphic readiness)

Maintain logits η_p and update them via delayed memory features $M(t)$:

$$\|\dot{\eta}_p(t) - \kappa_p G_p(M(t-\tau), M(t-\tau-\Delta t))\| \leq w_p = \mathrm{softmax}(\eta)_p.$$

Here $M(t)$ can be derived from $\Phi(\tilde{\rho}(t))$ or directly from sector-reduced signals.

A stabilizing choice is to make G Lipschitz-bounded and optionally project η to a bounded region.

6. QFT Constructions

We formalize three compatible QFT-grade extensions that arise naturally from the PMDM structure.

Throughout, let ϕ denote a bosonic scalar field for examples; extensions to fermions and gauge fields are straightforward.

(A) Mixture-of-QFTs: prime-sector generating functional

A.1 Definition

Assign each prime p a QFT action $S_p[\phi]$ (e.g., different couplings, masses, lattice spacing, boundary conditions).

Define the prime-mixture generating functional:

$$Z_{\mathrm{PM}}[J] = \sum_{p \in \mathcal{P}_N} w_p Z_p[J], \quad Z_p[J] = \int \mathcal{D}\phi \exp \left(-S_p[\phi] + \int d^d x J(x) \phi(x) \right).$$

This is the path-integral analog of diagonal PMDM: $\tilde{\rho} = \sum_p w_p |p\rangle \langle p| \otimes \rho^{(p)}$.

A.2 Example: Euclidean scalar field with prime-dependent mass

Let

$$S_p[\phi] = \int d^d x \left[\frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} m_p^2 \phi^2 + \frac{\lambda}{4!} \phi^4 \right], \quad m_p = m_0 p^{-\gamma}.$$

Then

$$Z_{\mathrm{PM}}[J] = \sum_p w_p Z(m_p; J).$$

Observable prediction: susceptibilities or correlation lengths become *mixtures* over m_p , leading to broadened transitions as w_p is tuned (e.g., by a “prime temperature” s in $w_p \propto p^{-s}$).

A.3 Relation to density matrices

At finite temperature β^{-1} , each sector defines $\rho_p \propto e^{-\beta H_p}$. The mixture $\rho = \sum_p w_p \rho_p$ matches the path-integral mixture above.

(B) Prime-graded field content with cross-prime couplers

B.1 Definition

Introduce a single theory with **prime-graded fields** $\{\phi_p\}_{p \in \mathcal{P}_N}$ and an action

$$S[\{\phi_p\}] = \sum_p S_p[\phi_p] + \sum_{p < q} \lambda_{pq} \Omega_{pq}[\phi_p, \phi_q].$$

This corresponds to a coherent PMDM regime where cross-prime couplers generate off-diagonal blocks ρ_{pq} .

B.2 Example: coupled scalar sectors (nearest-prime coupling)

Take

$$S_p[\phi_p] = \int d^d x \left[\frac{1}{2} (\partial \phi_p)^2 + \frac{1}{2} m_p^2 \phi_p^2 + \frac{\lambda_p}{4!} \phi_p^4 \right],$$

and cross-prime coupler

$$\Omega_{pq}[\phi_p, \phi_q] = \int d^d x (\phi_p - \phi_q)^2, \quad \lambda_{pq} = \lambda_0 K(|\log p - \log q|).$$

Here K can be localized (couple only adjacent primes in $\log p$) or long-range.

Interpretation: primes index a multiscale ladder; K controls how strongly adjacent scales synchronize.

B.3 Noncommuting operator viewpoint

In canonical quantization, each prime sector has operators $\mathcal{A}_p$. Couplers introduce terms mixing sectors, so evolution operators from different primes no longer commute in general:

$$[H_p, H_q] \neq 0 \quad \text{once } H_{\text{couple}} \neq 0.$$

This realizes the MOC “prime-indexed noncommutation” as a physical interaction.

(C) Prime-graded Keldysh boundary condition (density-matrix-first QFT)

C.1 Closed-time-path setup

The Keldysh generating functional with an initial density operator $\tilde{\rho}(t_0)$ is

$$Z[J_+, J_-] = \mathrm{Tr} \left(U[J_+] \tilde{\rho}(t_0) U[J_-]^\dagger \right),$$

where $U[J_+]$ evolves along the forward contour and $U[J_-]^\dagger$ along the backward contour.

C.2 Prime-graded initial condition

Take $\tilde{\rho}(t_0)$ as a prime-graded PMDM on $\mathcal{H}_{\mathbb{P}} \otimes \mathcal{H}_{\mathrm{HG}}$. In the diagonal regime:

$$\tilde{\rho}(t_0) = \sum_p w_p |p\rangle\langle p| \otimes \rho^{(p)}(t_0).$$

Then

$$Z[J_+, J_-] = \sum_p w_p \mathrm{Tr} \left(U_p[J_+] \rho^{(p)}(t_0) U_p[J_-]^\dagger \right).$$

In the coherent regime (off-diagonal ρ_{pq}), cross-terms appear:

$$Z[J_+, J_-] = \sum_{\{p,q\}} \mathrm{Tr} \left(U_p[J_+] \rho_{\{pq\}}(t_0) U_q[J_-]^\dagger \right),$$

which is a direct operational signature of prime-coherence.

C.3 Example: prime-weighted thermal boundary states

Let $\rho^{(p)}(t_0) \propto e^{-\beta H_p}$. Then

$$\tilde{\rho}(t_0) = \sum_p w_p |p\rangle\langle p| \otimes \frac{e^{-\beta H_p}}{\mathrm{Tr}(e^{-\beta H_p})}.$$

This is a **prime-mixture of thermal boundary conditions**; tuning w_p reshapes nonequilibrium response without changing the contour dynamics.

7. Crosswalk: PMDM $\leftrightarrow$ MOC $\leftrightarrow$ QFT

- **MOC “prime-indexed move”** $\leftrightarrow$ operator families $\tilde{U}_{p,\alpha}$ (controlled) plus optional cross-prime couplers $\tilde{V}$.
- **Hypergraph state+relation dynamics** $\leftrightarrow \mathcal{H}_V \otimes \mathcal{H}_E$ and prime-indexed operations on $\mathcal{H}_{\mathrm{HG}}$.
- **Resonance validation** $\leftrightarrow R(\tilde{\rho}; \mathcal{T})$ built from prime-aware feature extraction Φ .

- **QFT-grade extension** $\leftrightarrow$ (A) mixture over actions, (B) graded fields with couplers, (C) Keldysh boundary conditions from $\tilde{\rho}(t_0)$.
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8. Minimal simulation-ready recipes

8.1 Minimal toy for (A)

- Choose primes $\{2, 3, 5, 7\}$, weights $w_p \propto p^{-s}$.
- Define $m_p = m_0 p^\gamma$ in a 0+1D (quantum mechanics) or 1+1D lattice scalar.
- Compute Z_p numerically and form Z_{PM} .
- Measure correlator mixture: $\langle \phi \phi \rangle_{\text{PM}} = \sum_p w_p \langle \phi \phi \rangle_p$.

8.2 Minimal toy for (B)

- Fields ϕ_p for $p \in \{2, 3, 5, 7\}$.
- Action $S = \sum_p S_p + \sum_{p < q} \lambda_{pq} \int (\phi_p - \phi_q)^2$.
- Choose $\lambda_{pq} = \lambda_0 \exp(-\alpha |\log p - \log q|)$.
- Study synchronization vs. desynchronization across primes.

8.3 Minimal toy for (C)

- Pick $\tilde{\rho}(t_0) = \sum_p w_p |p\rangle \langle p| \otimes \rho_p$ with $\rho_p \propto e^{-\beta H_p}$.
 - Implement Keldysh response to a quench and compare response functions as s (in w_p) varies.
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9. Practical notes and guardrails

1. **Prime truncation:** treat N as a model capacity parameter.
 2. **Coherence control:** allow $\rho_{pq} \neq 0$ only if you can bound cross-prime coupler strength; otherwise start diagonal.
 3. **Resonance gaming:** ensure R depends on stable, multi-feature constraints (not a single easily-exploited scalar).
 4. **Commutator-aware scheduling:** for split-step implementations, reorder prime updates to minimize $\sum \|\mathcal{L}_p, \mathcal{L}_q\|$ within a time step.
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10. What counts as “success” empirically

- **Stability:** PSD and trace preserved; bounded trajectories under delays.
- **Robustness:** graceful degradation when primes are dropped or perturbed.
- **Expressivity:** coherent regime yields distinct predictions (cross terms) beyond convex mixtures.

- **QFT signatures:** mixture broadening (A), cross-prime synchronization/phase locking (B), boundary-condition-driven response reshaping (C).

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